Operating Systems and Concurrency

Concurrency 5

COMP2007

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Goals

Today

- Parallel dining philosophers
- Readers/writers problem

The Dining Philosophers Problem

Description

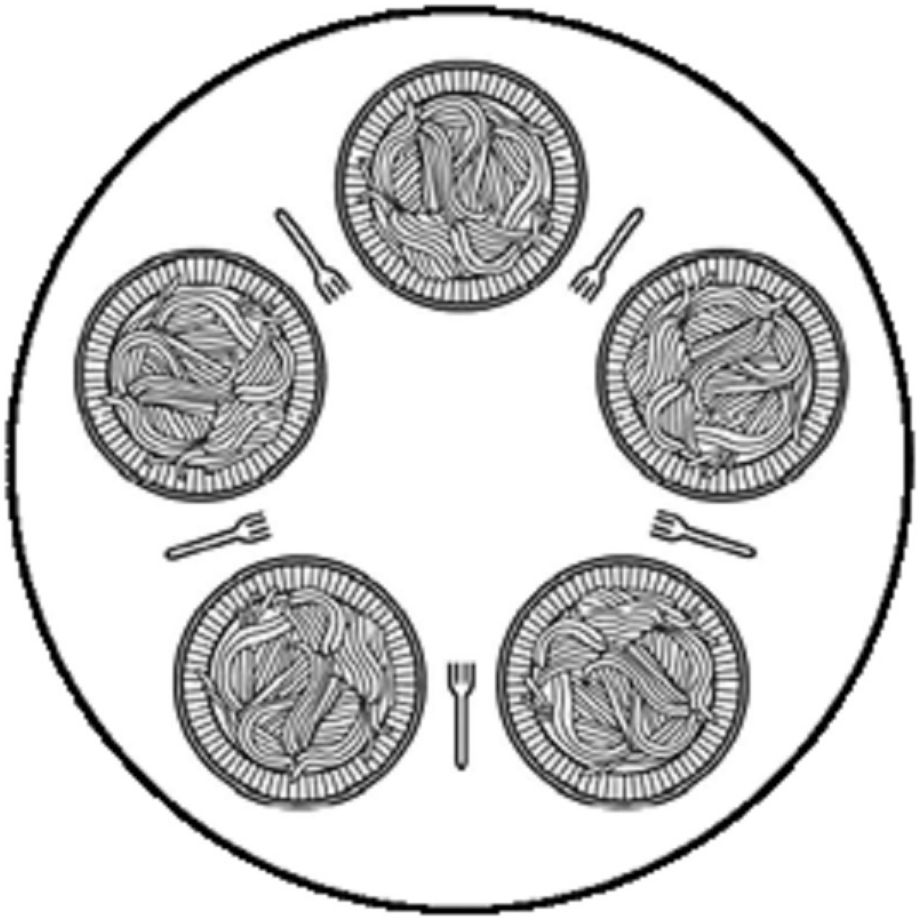


Figure: Tanenbaum, 4th edition

The Dining Philosophers Problem

Solutions 2: Global Mutex/Semaphore

```
sem_t eating;

void* philosopher(void* id) {
  int i = *((int*) id);
  int left = (i + N - 1) % N;
  int right = i % N;
  while(1)
  {
    printf("%d is thinking\n", i);
    printf("%d is hungry\n", i);
    sem_wait(&eating);       /**** mutex/semaphore ****/
    sem_wait(&forks[left]);
    sem_wait(&forks[right]);
    printf("%d is eating\n", i);
    sem_post(&forks[left]);
    sem_post(&forks[right]);
    sem_post(&eating);       /**** mutex/semaphore ****/
  }
}
```

The Dining Philosophers Problem

Solutions 2: Illustration

Philosopher 1	Philosopher 2	Philosopher 3	Philosopher 4	Philosopher 5
wait(&eating) 1=>0	wait(&eating)	wait(&eating)	wait(&eating)	wait(&eating)
wait(&forks[4])	wait(&forks[0])	wait(&forks[1])	wait(&forks[2])	wait(&forks[3])
wait(&forks[0])	wait(&forks[1])	wait(&forks[2])	wait(&forks[3])	wait(&forks[4])
...	...	...	...	...
// eating	// eating	// eating	// eating	// eating
...	...	...	...	...
post(&forks[4])	post(&forks[0])	post(&forks[1])	post(&forks[2])	post(&forks[3])
post(&forks[0])	post(&forks[1])	post(&forks[2])	post(&forks[3])	post(&forks[4])
post(&eating)	post(&eating)	post(&eating)	post(&eating)	post(&eating)

The Dining Philosophers Problem

Solutions 2: Illustration

```
Philosopher 1          Philosopher 2          Philosopher 3          Philosopher 4          Philosopher 5

wait(&eating)          wait(&eating)          wait(&eating)          wait(&eating)          wait(&eating)
wait(&forks[4]) 1=>0   wait(&forks[0])        wait(&forks[1])        wait(&forks[2])        wait(&forks[3])
wait(&forks[0])        wait(&forks[1])        wait(&forks[2])        wait(&forks[3])        wait(&forks[4])
...                    ...                    ...                    ...                    ...
// eating              // eating              // eating              // eating              // eating
...                    ...                    ...                    ...                    ...
post(&forks[4])        post(&forks[0])        post(&forks[1])        post(&forks[2])        post(&forks[3])
post(&forks[0])        post(&forks[1])        post(&forks[2])        post(&forks[3])        post(&forks[4])
post(&eating)          post(&eating)          post(&eating)          post(&eating)          post(&eating)
```

The Dining Philosophers Problem

Solutions 2: Illustration

```
Philosopher 1          Philosopher 2          Philosopher 3          Philosopher 4          Philosopher 5

wait(&eating)          wait(&eating)          wait(&eating)          wait(&eating)          wait(&eating)
wait(&forks[4])        wait(&forks[0])        wait(&forks[1])        wait(&forks[2])        wait(&forks[3])
wait(&forks[0]) 1=>0   wait(&forks[1])        wait(&forks[2])        wait(&forks[3])        wait(&forks[4])
...                    ...                    ...                    ...                    ...
// eating              // eating              // eating              // eating              // eating
...                    ...                    ...                    ...                    ...
post(&forks[4])        post(&forks[0])        post(&forks[1])        post(&forks[2])        post(&forks[3])
post(&forks[0])        post(&forks[1])        post(&forks[2])        post(&forks[3])        post(&forks[4])
post(&eating)          post(&eating)          post(&eating)          post(&eating)          post(&eating)
```

The Dining Philosophers Problem

Solutions 2: Illustration

Philosopher 1	Philosopher 2	Philosopher 3	Philosopher 4	Philosopher 5
wait(&eating)	**wait(&eating) 0=>-1**	wait(&eating)	wait(&eating)	wait(&eating)
wait(&forks[4])	wait(&forks[0])	wait(&forks[1])	wait(&forks[2])	wait(&forks[3])
wait(&forks[0])	wait(&forks[1])	wait(&forks[2])	wait(&forks[3])	wait(&forks[4])
...	...	...	...	...
// eating	// eating	// eating	// eating	// eating
...	...	...	...	...
post(&forks[4])	post(&forks[0])	post(&forks[1])	post(&forks[2])	post(&forks[3])
post(&forks[0])	post(&forks[1])	post(&forks[2])	post(&forks[3])	post(&forks[4])
post(&eating)	post(&eating)	post(&eating)	post(&eating)	post(&eating)

The Dining Philosophers Problem

Solutions 2: Illustration

```
Philosopher 1        Philosopher 2        Philosopher 3        Philosopher 4        Philosopher 5

wait(&eating)        wait(&eating)        wait(&eating) -1=>-2 wait(&eating)        wait(&eating)
wait(&forks[4])      wait(&forks[0])      wait(&forks[1])      wait(&forks[2])      wait(&forks[3])
wait(&forks[0])      wait(&forks[1])      wait(&forks[2])      wait(&forks[3])      wait(&forks[4])
...                  ...                  ...                  ...                  ...
// eating            // eating            // eating            // eating            // eating
...                  ...                  ...                  ...                  ...
post(&forks[4])      post(&forks[0])      post(&forks[1])      post(&forks[2])      post(&forks[3])
post(&forks[0])      post(&forks[1])      post(&forks[2])      post(&forks[3])      post(&forks[4])
post(&eating)        post(&eating)        post(&eating)        post(&eating)        post(&eating)
```

The Dining Philosophers Problem

Solutions 2: Illustration

```
Philosopher 1        Philosopher 2        Philosopher 3        Philosopher 4        Philosopher 5

wait(&eating)        wait(&eating)        wait(&eating)        wait(&eating) -2=>-3 wait(&eating)
wait(&forks[4])      wait(&forks[0])      wait(&forks[1])      wait(&forks[2])      wait(&forks[3])
wait(&forks[0])      wait(&forks[1])      wait(&forks[2])      wait(&forks[3])      wait(&forks[4])
...                  ...                  ...                  ...                  ...
// eating            // eating            // eating            // eating            // eating
...                  ...                  ...                  ...                  ...
post(&forks[4])      post(&forks[0])      post(&forks[1])      post(&forks[2])      post(&forks[3])
post(&forks[0])      post(&forks[1])      post(&forks[2])      post(&forks[3])      post(&forks[4])
post(&eating)        post(&eating)        post(&eating)        post(&eating)        post(&eating)
```

The Dining Philosophers Problem

Solutions 2: Illustration

```
Philosopher 1        Philosopher 2        Philosopher 3        Philosopher 4        Philosopher 5

wait(&eating)        wait(&eating)        wait(&eating)        wait(&eating)        wait(&eating) -3=>-4
wait(&forks[4])      wait(&forks[0])      wait(&forks[1])      wait(&forks[2])      wait(&forks[3])
wait(&forks[0])      wait(&forks[1])      wait(&forks[2])      wait(&forks[3])      wait(&forks[4])
...                  ...                  ...                  ...                  ...
// eating            // eating            // eating            // eating            // eating
...                  ...                  ...                  ...                  ...
post(&forks[4])      post(&forks[0])      post(&forks[1])      post(&forks[2])      post(&forks[3])
post(&forks[0])      post(&forks[1])      post(&forks[2])      post(&forks[3])      post(&forks[4])
post(&eating)        post(&eating)        post(&eating)        post(&eating)        post(&eating)
```

The Dining Philosophers Problem

Solutions 2: Illustration

```
Philosopher 1          Philosopher 2          Philosopher 3          Philosopher 4          Philosopher 5

wait(&eating)          wait(&eating)          wait(&eating)          wait(&eating)          wait(&eating)
wait(&forks[4])        wait(&forks[0])        wait(&forks[1])        wait(&forks[2])        wait(&forks[3])
wait(&forks[0])        wait(&forks[1])        wait(&forks[2])        wait(&forks[3])        wait(&forks[4])
...                    ...                    ...                    ...                    ...
// eating              // eating              // eating              // eating              // eating
...                    ...                    ...                    ...                    ...
post(&forks[4])  0=>1  post(&forks[0])        post(&forks[1])        post(&forks[2])        post(&forks[3])
post(&forks[0])        post(&forks[1])        post(&forks[2])        post(&forks[3])        post(&forks[4])
post(&eating)          post(&eating)          post(&eating)          post(&eating)          post(&eating)
```

The Dining Philosophers Problem

Solutions 2: Illustration

```
Philosopher 1          Philosopher 2          Philosopher 3          Philosopher 4          Philosopher 5

wait(&eating)          wait(&eating)          wait(&eating)          wait(&eating)          wait(&eating)
wait(&forks[4])        wait(&forks[0])        wait(&forks[1])        wait(&forks[2])        wait(&forks[3])
wait(&forks[0])        wait(&forks[1])        wait(&forks[2])        wait(&forks[3])        wait(&forks[4])
...                    ...                    ...                    ...                    ...
// eating              // eating              // eating              // eating              // eating
...                    ...                    ...                    ...                    ...
post(&forks[4])        post(&forks[0])        post(&forks[1])        post(&forks[2])        post(&forks[3])
post(&forks[0]) 0=>1   post(&forks[1])        post(&forks[2])        post(&forks[3])        post(&forks[4])
post(&eating)          post(&eating)          post(&eating)          post(&eating)          post(&eating)
```

The Dining Philosophers Problem

Solutions 2: Illustration

```
Philosopher 1          Philosopher 2          Philosopher 3          Philosopher 4          Philosopher 5

wait(&eating)          wait(&eating) (wake)   wait(&eating)          wait(&eating)          wait(&eating)
wait(&forks[4])        wait(&forks[0])        wait(&forks[1])        wait(&forks[2])        wait(&forks[3])
wait(&forks[0])        wait(&forks[1])        wait(&forks[2])        wait(&forks[3])        wait(&forks[4])
...                    ...                    ...                    ...                    ...
// eating              // eating              // eating              // eating              // eating
...                    ...                    ...                    ...                    ...
post(&forks[4])        post(&forks[0])        post(&forks[1])        post(&forks[2])        post(&forks[3])
post(&forks[0])        post(&forks[1])        post(&forks[2])        post(&forks[3])        post(&forks[4])
post(&eating) -4=>-3   post(&eating)          post(&eating)          post(&eating)          post(&eating)
```

The Dining Philosophers Problem

Last Time

Question in a Previous Year:

"Can I initialise the value of the `eating` *semaphore to 2 to create maximum parallelism"*

The Dining Philosophers Problem

Solutions 2: Global Mutex/Semaphore

```
sem_t eating;

void* philosopher(void* id) {
  int i = *((int*) id);
  int left = (i + N - 1) % N;
  int right = i % N;
  while(1)
  {
    printf("%d is thinking\n", i);
    printf("%d is hungry\n", i);
    sem_wait(&eating);       /**** mutex/semaphore ****/
    sem_wait(&forks[left]);
    sem_wait(&forks[right]);
    printf("%d is eating\n", i);
    sem_post(&forks[left]);
    sem_post(&forks[right]);
    sem_post(&eating);       /**** mutex/semaphore ****/
  }
}
```

The Dining Philosophers Problem

Solutions 2: Illustration

```
Philosopher 1        Philosopher 2        Philosopher 3        Philosopher 4        Philosopher 5

wait(&eating)        wait(&eating)        wait(&eating)        wait(&eating)        wait(&eating)
wait(&forks[4])      wait(&forks[0])      wait(&forks[1])      wait(&forks[2])      wait(&forks[3])
wait(&forks[0])      wait(&forks[1])      wait(&forks[2])      wait(&forks[3])      wait(&forks[4])
...                  ...                  ...                  ...                  ...
// eating            // eating            // eating            // eating            // eating
...                  ...                  ...                  ...                  ...
post(&forks[4])      post(&forks[0])      post(&forks[1])      post(&forks[2])      post(&forks[3])
post(&forks[0])      post(&forks[1])      post(&forks[2])      post(&forks[3])      post(&forks[4])
post(&eating)        post(&eating)        post(&eating)        post(&eating)        post(&eating)
```

The Dining Philosophers Problem

Solutions 2: Illustration

Philosopher 1	Philosopher 2	Philosopher 3	Philosopher 4	Philosopher 5
wait(&eating)	**wait(&eating)2=>1**	wait(&eating)	wait(&eating)	wait(&eating)
wait(&forks[4])	wait(&forks[0])	wait(&forks[1])	wait(&forks[2])	wait(&forks[3])
wait(&forks[0])	wait(&forks[1])	wait(&forks[2])	wait(&forks[3])	wait(&forks[4])
...	...	...	...	...
// eating	// eating	// eating	// eating	// eating
...	...	...	...	...
post(&forks[4])	post(&forks[0])	post(&forks[1])	post(&forks[2])	post(&forks[3])
post(&forks[0])	post(&forks[1])	post(&forks[2])	post(&forks[3])	post(&forks[4])
post(&eating)	post(&eating)	post(&eating)	post(&eating)	post(&eating)

The Dining Philosophers Problem

Solutions 2: Illustration

Philosopher 1	Philosopher 2	Philosopher 3	Philosopher 4	Philosopher 5
wait(&eating)	wait(&eating)	wait(&eating)	wait(&eating)	wait(&eating)
wait(&forks[4])	**wait(&forks[0])1=>0**	wait(&forks[1])	wait(&forks[2])	wait(&forks[3])
wait(&forks[0])	wait(&forks[1])	wait(&forks[2])	wait(&forks[3])	wait(&forks[4])
...	...	...	...	...
// eating	// eating	// eating	// eating	// eating
...	...	...	...	...
post(&forks[4])	post(&forks[0])	post(&forks[1])	post(&forks[2])	post(&forks[3])
post(&forks[0])	post(&forks[1])	post(&forks[2])	post(&forks[3])	post(&forks[4])
post(&eating)	post(&eating)	post(&eating)	post(&eating)	post(&eating)

The Dining Philosophers Problem

Solutions 2: Illustration

```
Philosopher 1        Philosopher 2        Philosopher 3        Philosopher 4        Philosopher 5

wait(&eating)        wait(&eating)        wait(&eating)        wait(&eating)        wait(&eating)
wait(&forks[4])      wait(&forks[0])      wait(&forks[1])      wait(&forks[2])      wait(&forks[3])
wait(&forks[0])      wait(&forks[1])1=>0  wait(&forks[2])      wait(&forks[3])      wait(&forks[4])
...                  ...                  ...                  ...                  ...
// eating            // eating            // eating            // eating            // eating
...                  ...                  ...                  ...                  ...
post(&forks[4])      post(&forks[0])      post(&forks[1])      post(&forks[2])      post(&forks[3])
post(&forks[0])      post(&forks[1])      post(&forks[2])      post(&forks[3])      post(&forks[4])
post(&eating)        post(&eating)        post(&eating)        post(&eating)        post(&eating)
```

The Dining Philosophers Problem

Solutions 2: Illustration

Philosopher 1	Philosopher 2	Philosopher 3	Philosopher 4	Philosopher 5
wait(&eating)	wait(&eating)	**wait(&eating)1=>0**	wait(&eating)	wait(&eating)
wait(&forks[4])	wait(&forks[0])	wait(&forks[1])	wait(&forks[2])	wait(&forks[3])
wait(&forks[0])	wait(&forks[1])	wait(&forks[2])	wait(&forks[3])	wait(&forks[4])
...	...	...	...	...
// eating	// eating	// eating	// eating	// eating
...	...	...	...	...
post(&forks[4])	post(&forks[0])	post(&forks[1])	post(&forks[2])	post(&forks[3])
post(&forks[0])	post(&forks[1])	post(&forks[2])	post(&forks[3])	post(&forks[4])
post(&eating)	post(&eating)	post(&eating)	post(&eating)	post(&eating)

The Dining Philosophers Problem

Solutions 2: Illustration

Philosopher 1	Philosopher 2	Philosopher 3	Philosopher 4	Philosopher 5
wait(&eating)	wait(&eating)	wait(&eating)	wait(&eating)	wait(&eating)
wait(&forks[4])	wait(&forks[0])	**wait(&forks[1])0=>-1**	wait(&forks[2])	wait(&forks[3])
wait(&forks[0])	**wait(&forks[1])**	wait(&forks[2])	wait(&forks[3])	wait(&forks[4])
...	...	...	...	...
// eating	// eating	// eating	// eating	// eating
...	...	...	...	...
post(&forks[4])	post(&forks[0])	post(&forks[1])	post(&forks[2])	post(&forks[3])
post(&forks[0])	post(&forks[1])	post(&forks[2])	post(&forks[3])	post(&forks[4])
post(&eating)	post(&eating)	post(&eating)	post(&eating)	post(&eating)

The Dining Philosophers Problem

Solutions 2: Illustration

Philosopher 1	Philosopher 2	Philosopher 3	Philosopher 4	Philosopher 5
wait(&eating)	wait(&eating)	wait(&eating)	**wait(&eating)0=>-1**	wait(&eating)
wait(&forks[4])	wait(&forks[0])	wait(&forks[1])	wait(&forks[2])	wait(&forks[3])
wait(&forks[0])	wait(&forks[1])	wait(&forks[2])	wait(&forks[3])	wait(&forks[4])
...	...	...	...	...
// eating	// eating	// eating	// eating	// eating
...	...	...	...	...
post(&forks[4])	post(&forks[0])	post(&forks[1])	post(&forks[2])	post(&forks[3])
post(&forks[0])	post(&forks[1])	post(&forks[2])	post(&forks[3])	post(&forks[4])
post(&eating)	post(&eating)	post(&eating)	post(&eating)	post(&eating)

The Dining Philosophers Problem

Solutions 2: Illustration

Philosopher 1	Philosopher 2	Philosopher 3	Philosopher 4	Philosopher 5
wait(&eating)	wait(&eating)	wait(&eating)	wait(&eating)	**wait(&eating)-1=>-2**
wait(&forks[4])	wait(&forks[0])	wait(&forks[1])	wait(&forks[2])	wait(&forks[3])
wait(&forks[0])	wait(&forks[1])	wait(&forks[2])	wait(&forks[3])	wait(&forks[4])
...	...	...	...	...
// eating	// eating	// eating	// eating	// eating
...	...	...	...	...
post(&forks[4])	post(&forks[0])	post(&forks[1])	post(&forks[2])	post(&forks[3])
post(&forks[0])	post(&forks[1])	post(&forks[2])	post(&forks[3])	post(&forks[4])
post(&eating)	post(&eating)	post(&eating)	post(&eating)	post(&eating)

The Dining Philosophers Problem

Solutions 2: Illustration

Philosopher 1	Philosopher 2	Philosopher 3	Philosopher 4	Philosopher 5
wait(&eating)-2=>-3	wait(&eating)	wait(&eating)	wait(&eating)	wait(&eating)
wait(&forks[4])	wait(&forks[0])	wait(&forks[1])	wait(&forks[2])	wait(&forks[3])
wait(&forks[0])	wait(&forks[1])	wait(&forks[2])	wait(&forks[3])	wait(&forks[4])
...	...	...	...	...
// eating	// eating	// eating	// eating	// eating
...	...	...	...	...
post(&forks[4])	post(&forks[0])	post(&forks[1])	post(&forks[2])	post(&forks[3])
post(&forks[0])	post(&forks[1])	post(&forks[2])	post(&forks[3])	post(&forks[4])
post(&eating)	post(&eating)	post(&eating)	post(&eating)	post(&eating)

The Dining Philosophers Problem

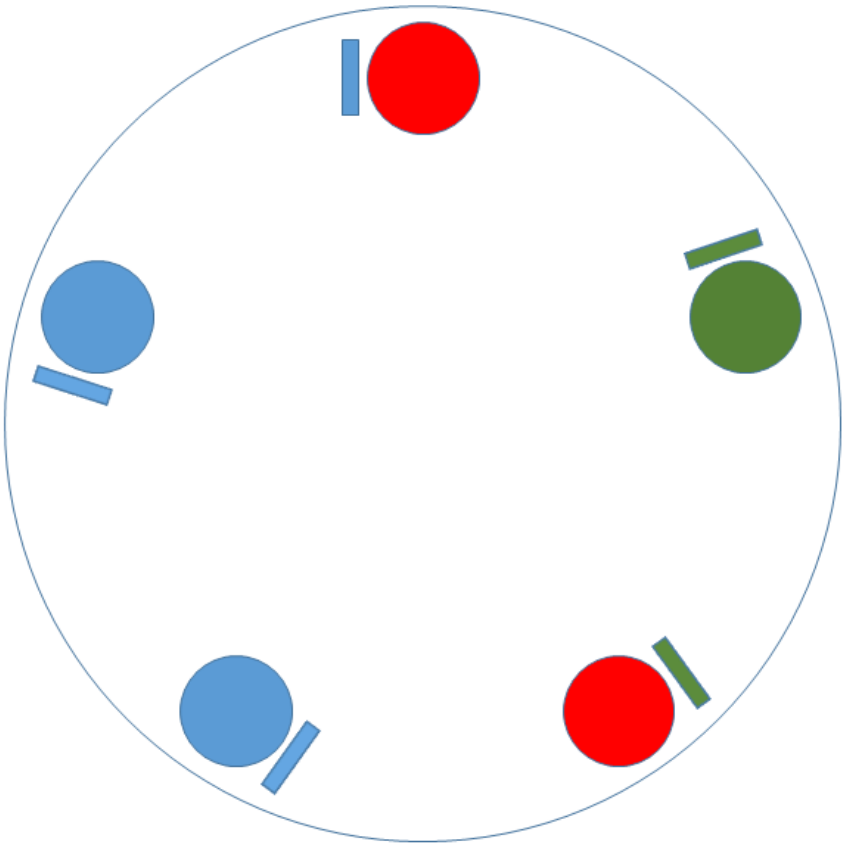
Solution 3: Maximum Parallelism

- A **more sophisticated solution** is necessary to allow **maximum parallelism**
- The solution uses:
  - `state[N]`: one **state variable** for every philosopher (`THINKING`, `HUNGRY`, `EATING`)
  - `phil[N]`: one **semaphore per *philosopher*** rather than per **fork**.
    - The philosopher **blocks** if one of their neighbours are eating
    - The **neighbours wake up the philosopher** if they have finished eating
  - `sync`: one **semaphore/mutex** to enforce **mutual exclusion** of the critical section while updating the **states**.

The Dining Philosophers Problem

Solution 3: Maximum Parallelism

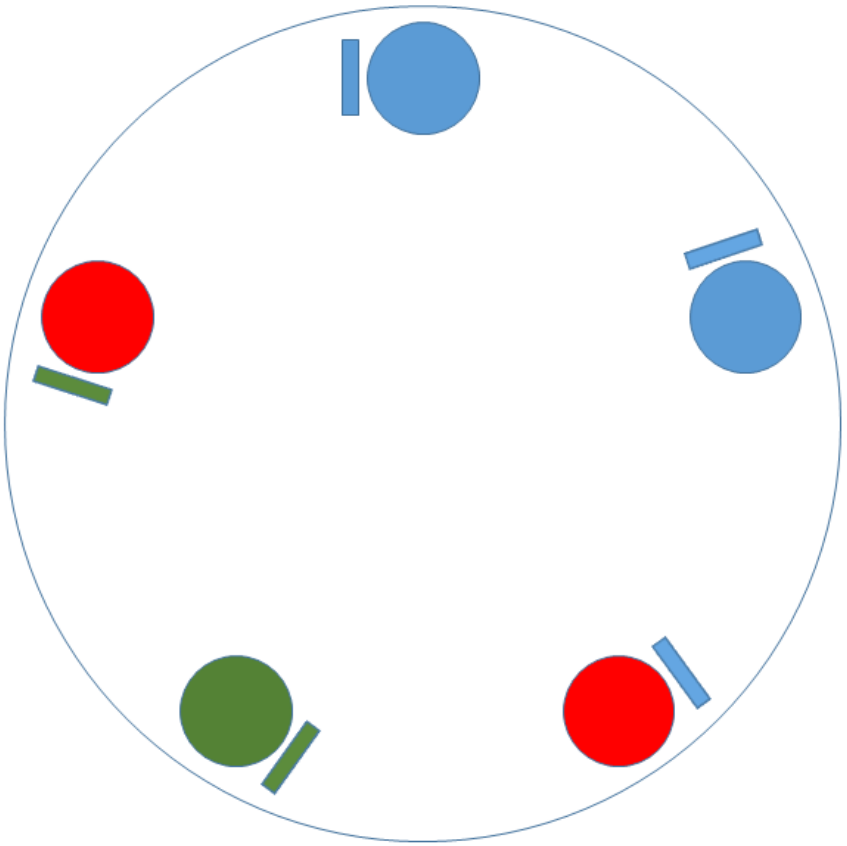
- A philosopher can only **start eating** if their **neighbours are not eating**



The Dining Philosophers Problem

Solution 3: Maximum Parallelism

- A philosopher can only **start eating** if their **neighbours are not eating**



The Dining Philosophers Problem

Solution 3: Maximum Parallelism

```
#define N 5
#define THINKING 1
#define HUNGRY 2
#define EATING 3

int state[N] = {THINKING, THINKING, THINKING, THINKING, THINKING};
sem_t phil[N];  // Initial capacity 0
sem_t sync;     // Initial capacity 1

void* philosopher(void* id) {
  int i = *((int *) id);
  while(1) {
    printf("%d is thinking\n", i);
    take_forks(i);
    printf("%d is eating\n", i);
    put_forks(i);
  }
}
```

The Dining Philosophers Problem

Solution 3: Maximum Parallelism

```
void test(int i) {
  int left = (i + N - 1) % N;
  int right = (i + 1) % N;
  if(state[i] == HUNGRY
    && state[left] != EATING
    && state[right] != EATING) {
    state[i] = EATING;
    sem_post(&phil[i]);
  }
}
```

The Dining Philosophers Problem

Solution 3: Maximum Parallelism

```
void take_forks(int i) {
  sem_wait(&sync);
  state[i] = HUNGRY;
  test(i);
  sem_post(&sync);
  sem_wait(&phil[i]);
}

void put_forks(int i) {
  int left = (i + N - 1) % N;
  int right = (i + 1) % N;
  sem_wait(&sync);
  state[i] = THINKING;
  test(left);
  test(right);
  sem_post(&sync);
}
```

```
void test(int i) {
  int left = (i + N - 1) % N;
  int right = (i + 1) % N;
  if(state[i] == HUNGRY
    && state[left] != EATING
    && state[right] != EATING) {
    state[i] = EATING;
    sem_post(&phil[i]);
  }
}
```

The Dining Philosophers Problem

Solution 3: Maximum Parallelism

```
Philosopher 2
(left = 1, right = 3)

wait(&sync)
state[2]=HUNGRY
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
post(&sync)
wait(&phil[2])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[2] = THINK
// test neighbours
if(state[1]==HUNGRY
  && state[5]!=EAT
  && state[2]!=EAT){
  state[1]=EAT
  post(&phil[1])
}
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
```

```
Philosopher 3
(left = 2, right = 4)

wait(&sync)  // 1 => 0
state[3]=HUNGRY
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
wait(&phil[3])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[3] = THINK
// test neighbours
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
```

```
Philosopher 4
(left = 3, right = 5)

wait(&sync)
state[4]=HUNGRY
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
wait(&phil[4])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[4] = THINK
// test neighbours
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
if(state[5]==HUNGRY
  && state[4]!=EAT
  && state[1]!=EAT){
  state[5]=EAT
  post(&phil[5])
}
post(&sync)
```

The Dining Philosophers Problem

Solution 3: Maximum Parallelism

```
Philosopher 2
(left = 1, right = 3)

wait(&sync)
state[2]=HUNGRY
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
post(&sync)
wait(&phil[2])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[2] = THINK
// test neighbours
if(state[1]==HUNGRY
  && state[5]!=EAT
  && state[2]!=EAT){
  state[1]=EAT
  post(&phil[1])
}
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
```

```
Philosopher 3
(left = 2, right = 4)

wait(&sync)
state[3]=HUNGRY
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
wait(&phil[3])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[3] = THINK
// test neighbours
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
```

```
Philosopher 4
(left = 3, right = 5)

wait(&sync)
state[4]=HUNGRY
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
wait(&phil[4])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[4] = THINK
// test neighbours
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
if(state[5]==HUNGRY
  && state[4]!=EAT
  && state[1]!=EAT){
  state[5]=EAT
  post(&phil[5])
}
post(&sync)
```

The Dining Philosophers Problem

Solution 3: Maximum Parallelism

```
Philosopher 2
(left = 1, right = 3)

wait(&sync)
state[2]=HUNGRY
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
post(&sync)
wait(&phil[2])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[2] = THINK
// test neighbours
if(state[1]==HUNGRY
  && state[5]!=EAT
  && state[2]!=EAT){
  state[1]=EAT
  post(&phil[1])
}
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
```

```
Philosopher 3
(left = 2, right = 4)

wait(&sync)
state[3]=HUNGRY
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])  // 0 => 1
}
post(&sync)
wait(&phil[3])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[3] = THINK
// test neighbours
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
```

```
Philosopher 4
(left = 3, right = 5)

wait(&sync)
state[4]=HUNGRY
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
wait(&phil[4])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[4] = THINK
// test neighbours
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
if(state[5]==HUNGRY
  && state[4]!=EAT
  && state[1]!=EAT){
  state[5]=EAT
  post(&phil[5])
}
post(&sync)
```

The Dining Philosophers Problem

Solution 3: Maximum Parallelism

```
Philosopher 2
(left = 1, right = 3)

wait(&sync)
state[2]=HUNGRY
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
post(&sync)
wait(&phil[2])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[2] = THINK
// test neighbours
if(state[1]==HUNGRY
  && state[5]!=EAT
  && state[2]!=EAT){
  state[1]=EAT
  post(&phil[1])
}
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
```

```
Philosopher 3
(left = 2, right = 4)

wait(&sync)
state[3]=HUNGRY
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)  // 0 => 1
wait(&phil[3])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[3] = THINK
// test neighbours
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
```

```
Philosopher 4
(left = 3, right = 5)

wait(&sync)
state[4]=HUNGRY
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
wait(&phil[4])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[4] = THINK
// test neighbours
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
if(state[5]==HUNGRY
  && state[4]!=EAT
  && state[1]!=EAT){
  state[5]=EAT
  post(&phil[5])
}
post(&sync)
```

The Dining Philosophers Problem

Solution 3: Maximum Parallelism

```
Philosopher 2
(left = 1, right = 3)

wait(&sync)
state[2]=HUNGRY
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
post(&sync)
wait(&phil[2])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[2] = THINK
// test neighbours
if(state[1]==HUNGRY
  && state[5]!=EAT
  && state[2]!=EAT){
  state[1]=EAT
  post(&phil[1])
}
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
```

```
Philosopher 3
(left = 2, right = 4)

wait(&sync)
state[3]=HUNGRY
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])  // 0 => 1
}
post(&sync)
wait(&phil[3])  // 1 => 0

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[3] = THINK
// test neighbours
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
```

```
Philosopher 4
(left = 3, right = 5)

wait(&sync)
state[4]=HUNGRY
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
wait(&phil[4])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[4] = THINK
// test neighbours
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
if(state[5]==HUNGRY
  && state[4]!=EAT
  && state[1]!=EAT){
  state[5]=EAT
  post(&phil[5])
}
post(&sync)
```

The Dining Philosophers Problem

Solution 3: Maximum Parallelism

```
Philosopher 2
(left = 1, right = 3)

wait(&sync)
state[2]=HUNGRY
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
post(&sync)
wait(&phil[2])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[2] = THINK
// test neighbours
if(state[1]==HUNGRY
  && state[5]!=EAT
  && state[2]!=EAT){
  state[1]=EAT
  post(&phil[1])
}
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
```

```
Philosopher 3
(left = 2, right = 4)

wait(&sync)
state[3]=HUNGRY
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
wait(&phil[3])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[3] = THINK
// test neighbours
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
```

```
Philosopher 4
(left = 3, right = 5)

wait(&sync)
state[4]=HUNGRY
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
wait(&phil[4])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[4] = THINK
// test neighbours
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
if(state[5]==HUNGRY
  && state[4]!=EAT
  && state[1]!=EAT){
  state[5]=EAT
  post(&phil[5])
}
post(&sync)
```

The Dining Philosophers Problem

Solution 3: Maximum Parallelism

```
Philosopher 2
(left = 1, right = 3)

wait(&sync)
state[2]=HUNGRY
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
post(&sync)
wait(&phil[2])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[2] = THINK
// test neighbours
if(state[1]==HUNGRY
  && state[5]!=EAT
  && state[2]!=EAT){
  state[1]=EAT
  post(&phil[1])
}
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
```

```
Philosopher 3
(left = 2, right = 4)

wait(&sync)
state[3]=HUNGRY
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
wait(&phil[3])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)  // 1 => 0
state[3] = THINK
// test neighbours
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
```

```
Philosopher 4
(left = 3, right = 5)

wait(&sync)
state[4]=HUNGRY
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
wait(&phil[4])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[4] = THINK
// test neighbours
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
if(state[5]==HUNGRY
  && state[4]!=EAT
  && state[1]!=EAT){
  state[5]=EAT
  post(&phil[5])
}
post(&sync)
```

The Dining Philosophers Problem

Solution 3: Maximum Parallelism

```
Philosopher 2
(left = 1, right = 3)

wait(&sync)
state[2]=HUNGRY
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
post(&sync)
wait(&phil[2])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[2] = THINK
// test neighbours
if(state[1]==HUNGRY
  && state[5]!=EAT
  && state[2]!=EAT){
  state[1]=EAT
  post(&phil[1])
}
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
```

```
Philosopher 3
(left = 2, right = 4)

wait(&sync)
state[3]=HUNGRY
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
wait(&phil[3])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[3] = THINK
// test neighbours
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
```

```
Philosopher 4
(left = 3, right = 5)

wait(&sync)
state[4]=HUNGRY
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
wait(&phil[4])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[4] = THINK
// test neighbours
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
if(state[5]==HUNGRY
  && state[4]!=EAT
  && state[1]!=EAT){
  state[5]=EAT
  post(&phil[5])
}
post(&sync)
```

The Dining Philosophers Problem

Solution 3: Maximum Parallelism

```
Philosopher 2
(left = 1, right = 3)

wait(&sync)
state[2]=HUNGRY
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
post(&sync)
wait(&phil[2])  // assume == -1 (wakeup)

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[2] = THINK
// test neighbours
if(state[1]==HUNGRY
  && state[5]!=EAT
  && state[2]!=EAT){
  state[1]=EAT
  post(&phil[1])
}
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
```

```
Philosopher 3
(left = 2, right = 4)

wait(&sync)
state[3]=HUNGRY
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
wait(&phil[3])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[3] = THINK
// test neighbours
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])  // -1 => 0
}
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
```

```
Philosopher 4
(left = 3, right = 5)

wait(&sync)
state[4]=HUNGRY
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
wait(&phil[4])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[4] = THINK
// test neighbours
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
if(state[5]==HUNGRY
  && state[4]!=EAT
  && state[1]!=EAT){
  state[5]=EAT
  post(&phil[5])
}
post(&sync)
```

The Dining Philosophers Problem

Solution 3: Maximum Parallelism

```
Philosopher 2
(left = 1, right = 3)

wait(&sync)
state[2]=HUNGRY
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
post(&sync)
wait(&phil[2])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[2] = THINK
// test neighbours
if(state[1]==HUNGRY
  && state[5]!=EAT
  && state[2]!=EAT){
  state[1]=EAT
  post(&phil[1])
}
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
```

```
Philosopher 3
(left = 2, right = 4)

wait(&sync)
state[3]=HUNGRY
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
wait(&phil[3])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[3] = THINK
// test neighbours
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])  // -1 => 0
}
post(&sync)
```

```
Philosopher 4
(left = 3, right = 5)

wait(&sync)
state[4]=HUNGRY
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
wait(&phil[4]) // assume -1 (wakeup)

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[4] = THINK
// test neighbours
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
if(state[5]==HUNGRY
  && state[4]!=EAT
  && state[1]!=EAT){
  state[5]=EAT
  post(&phil[5])
}
post(&sync)
```

The Dining Philosophers Problem

Solution 3: Maximum Parallelism

```
Philosopher 2
(left = 1, right = 3)

wait(&sync)
state[2]=HUNGRY
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
post(&sync)
wait(&phil[2])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[2] = THINK
// test neighbours
if(state[1]==HUNGRY
  && state[5]!=EAT
  && state[2]!=EAT){
  state[1]=EAT
  post(&phil[1])
}
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
```

```
Philosopher 3
(left = 2, right = 4)

wait(&sync)
state[3]=HUNGRY
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
wait(&phil[3])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[3] = THINK
// test neighbours
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)  // 0 => 1
```

```
Philosopher 4
(left = 3, right = 5)

wait(&sync)
state[4]=HUNGRY
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
wait(&phil[4])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[4] = THINK
// test neighbours
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
if(state[5]==HUNGRY
  && state[4]!=EAT
  && state[1]!=EAT){
  state[5]=EAT
  post(&phil[5])
}
post(&sync)
```

The Dining Philosophers Problem

Solution 3: Maximum Parallelism

```
Philosopher 2
(left = 1, right = 3)

wait(&sync)
state[2]=HUNGRY
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
post(&sync)
wait(&phil[2])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[2] = THINK
// test neighbours
if(state[1]==HUNGRY
  && state[5]!=EAT
  && state[2]!=EAT){
  state[1]=EAT
  post(&phil[1])
}
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
```

```
Philosopher 3
(left = 2, right = 4)

wait(&sync)
state[3]=HUNGRY
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
wait(&phil[3])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[3] = THINK
// test neighbours
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
```

```
Philosopher 4
(left = 3, right = 5)

wait(&sync)
state[4]=HUNGRY
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
wait(&phil[4])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[4] = THINK
// test neighbours
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
if(state[5]==HUNGRY
  && state[4]!=EAT
  && state[1]!=EAT){
  state[5]=EAT
  post(&phil[5])
}
post(&sync)
```

The Dining Philosophers Problem

Solution 3: Maximum Parallelism

```
Philosopher 2
(left = 1, right = 3)

wait(&sync)
state[2]=HUNGRY
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
post(&sync)
wait(&phil[2])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[2] = THINK
// test neighbours
if(state[1]==HUNGRY
  && state[5]!=EAT
  && state[2]!=EAT){
  state[1]=EAT
  post(&phil[1])
}
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
```

```
Philosopher 3
(left = 2, right = 4)

wait(&sync)  // 1 => 0
state[3]=HUNGRY
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
wait(&phil[3])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[3] = THINK
// test neighbours
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
```

```
Philosopher 4
(left = 3, right = 5)

wait(&sync)
state[4]=HUNGRY
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
wait(&phil[4])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[4] = THINK
// test neighbours
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
if(state[5]==HUNGRY
  && state[4]!=EAT
  && state[1]!=EAT){
  state[5]=EAT
  post(&phil[5])
}
post(&sync)
```

The Dining Philosophers Problem

Solution 3: Maximum Parallelism

```
Philosopher 2
(left = 1, right = 3)

wait(&sync)
state[2]=HUNGRY
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
post(&sync)
wait(&phil[2])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[2] = THINK
// test neighbours
if(state[1]==HUNGRY
  && state[5]!=EAT
  && state[2]!=EAT){
  state[1]=EAT
  post(&phil[1])
}
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
```

```
Philosopher 3
(left = 2, right = 4)

wait(&sync)
state[3]=HUNGRY
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
wait(&phil[3])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[3] = THINK
// test neighbours
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
```

```
Philosopher 4
(left = 3, right = 5)

wait(&sync)
state[4]=HUNGRY
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
wait(&phil[4])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[4] = THINK
// test neighbours
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
if(state[5]==HUNGRY
  && state[4]!=EAT
  && state[1]!=EAT){
  state[5]=EAT
  post(&phil[5])
}
post(&sync)
```

The Dining Philosophers Problem

Solution 3: Maximum Parallelism

```
Philosopher 2
(left = 1, right = 3)

wait(&sync)
state[2]=HUNGRY
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
post(&sync)
wait(&phil[2])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[2] = THINK
// test neighbours
if(state[1]==HUNGRY
  && state[5]!=EAT
  && state[2]!=EAT){
  state[1]=EAT
  post(&phil[1])
}
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
```

```
Philosopher 3
(left = 2, right = 4)

wait(&sync)
state[3]=HUNGRY
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
wait(&phil[3])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[3] = THINK
// test neighbours
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
```

```
Philosopher 4
(left = 3, right = 5)

wait(&sync)
state[4]=HUNGRY
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
wait(&phil[4])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[4] = THINK
// test neighbours
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
if(state[5]==HUNGRY
  && state[4]!=EAT
  && state[1]!=EAT){
  state[5]=EAT
  post(&phil[5])
}
post(&sync)
```

The Dining Philosophers Problem

Solution 3: Maximum Parallelism

```
Philosopher 2
(left = 1, right = 3)

wait(&sync)
state[2]=HUNGRY
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
post(&sync)
wait(&phil[2])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[2] = THINK
// test neighbours
if(state[1]==HUNGRY
  && state[5]!=EAT
  && state[2]!=EAT){
  state[1]=EAT
  post(&phil[1])
}
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
```

```
Philosopher 3
(left = 2, right = 4)

wait(&sync)
state[3]=HUNGRY
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)  // 0 => 1
wait(&phil[3])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[3] = THINK
// test neighbours
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
```

```
Philosopher 4
(left = 3, right = 5)

wait(&sync)
state[4]=HUNGRY
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
wait(&phil[4])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[4] = THINK
// test neighbours
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
if(state[5]==HUNGRY
  && state[4]!=EAT
  && state[1]!=EAT){
  state[5]=EAT
  post(&phil[5])
}
post(&sync)
```

The Dining Philosophers Problem

Solution 3: Maximum Parallelism

```
Philosopher 2
(left = 1, right = 3)

wait(&sync)
state[2]=HUNGRY
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
post(&sync)
wait(&phil[2])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[2] = THINK
// test neighbours
if(state[1]==HUNGRY
  && state[5]!=EAT
  && state[2]!=EAT){
  state[1]=EAT
  post(&phil[1])
}
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
```

```
Philosopher 3
(left = 2, right = 4)

wait(&sync)
state[3]=HUNGRY
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
wait(&phil[3])  // 0 => -1 (sleeping)

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[3] = THINK
// test neighbours
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
```

```
Philosopher 4
(left = 3, right = 5)

wait(&sync)
state[4]=HUNGRY
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
wait(&phil[4])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[4] = THINK
// test neighbours
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
if(state[5]==HUNGRY
  && state[4]!=EAT
  && state[1]!=EAT){
  state[5]=EAT
  post(&phil[5])
}
post(&sync)
```

The Dining Philosophers Problem

Solution 3: Maximum Parallelism

```
Philosopher 2
(left = 1, right = 3)

wait(&sync)
state[2]=HUNGRY
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
post(&sync)
wait(&phil[2])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)  // 1 => 0
state[2] = THINK
// test neighbours
if(state[1]==HUNGRY
  && state[5]!=EAT
  && state[2]!=EAT){
  state[1]=EAT
  post(&phil[1])
}
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
```

```
Philosopher 3
(left = 2, right = 4)

wait(&sync)
state[3]=HUNGRY
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
wait(&phil[3])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[3] = THINK
// test neighbours
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
```

```
Philosopher 4
(left = 3, right = 5)

wait(&sync)
state[4]=HUNGRY
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
wait(&phil[4])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[4] = THINK
// test neighbours
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
if(state[5]==HUNGRY
  && state[4]!=EAT
  && state[1]!=EAT){
  state[5]=EAT
  post(&phil[5])
}
post(&sync)
```

The Dining Philosophers Problem

Solution 3: Maximum Parallelism

```
Philosopher 2
(left = 1, right = 3)

wait(&sync)
state[2]=HUNGRY
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
post(&sync)
wait(&phil[2])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[2] = THINK
// test neighbours
if(state[1]==HUNGRY
  && state[5]!=EAT
  && state[2]!=EAT){
  state[1]=EAT
  post(&phil[1])
}
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
```

```
Philosopher 3
(left = 2, right = 4)

wait(&sync)
state[3]=HUNGRY
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
wait(&phil[3])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[3] = THINK
// test neighbours
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
```

```
Philosopher 4
(left = 3, right = 5)

wait(&sync)
state[4]=HUNGRY
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
wait(&phil[4])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[4] = THINK
// test neighbours
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
if(state[5]==HUNGRY
  && state[4]!=EAT
  && state[1]!=EAT){
  state[5]=EAT
  post(&phil[5])
}
post(&sync)
```

The Dining Philosophers Problem

Solution 3: Maximum Parallelism

```
Philosopher 2
(left = 1, right = 3)

wait(&sync)
state[2]=HUNGRY
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
post(&sync)
wait(&phil[2])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[2] = THINK
// test neighbours
if(state[1]==HUNGRY
  && state[5]!=EAT
  && state[2]!=EAT){
  state[1]=EAT
  post(&phil[1])
}
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
```

```
Philosopher 3
(left = 2, right = 4)

wait(&sync)
state[3]=HUNGRY
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
wait(&phil[3])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[3] = THINK
// test neighbours
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
```

```
Philosopher 4
(left = 3, right = 5)

wait(&sync)
state[4]=HUNGRY
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
wait(&phil[4])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[4] = THINK
// test neighbours
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
if(state[5]==HUNGRY
  && state[4]!=EAT
  && state[1]!=EAT){
  state[5]=EAT
  post(&phil[5])
}
post(&sync)
```

The Dining Philosophers Problem

Solution 3: Maximum Parallelism

```
Philosopher 2
(left = 1, right = 3)

wait(&sync)
state[2]=HUNGRY
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
post(&sync)
wait(&phil[2])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[2] = THINK
// test neighbours
if(state[1]==HUNGRY
  && state[5]!=EAT
  && state[2]!=EAT){
  state[1]=EAT
  post(&phil[1])
}
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])  // -1 => 0
}
post(&sync)
```

```
Philosopher 3
(left = 2, right = 4)

wait(&sync)
state[3]=HUNGRY
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
post(&sync)
wait(&phil[3])  // wakeup

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[3] = THINK
// test neighbours
if(state[2]==HUNGRY
  && state[1]!=EAT
  && state[3]!=EAT){
  state[2]=EAT
  post(&phil[2])
}
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
```

```
Philosopher 4
(left = 3, right = 5)

wait(&sync)
state[4]=HUNGRY
if(state[4]==HUNGRY
  && state[3]!=EAT
  && state[5]!=EAT){
  state[4]=EAT
  post(&phil[4])
}
post(&sync)
wait(&phil[4])

// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT
// EAT EAT EAT EAT EAT

wait(&sync)
state[4] = THINK
// test neighbours
if(state[3]==HUNGRY
  && state[2]!=EAT
  && state[4]!=EAT){
  state[3]=EAT
  post(&phil[3])
}
if(state[5]==HUNGRY
  && state[4]!=EAT
  && state[1]!=EAT){
  state[5]=EAT
  post(&phil[5])
}
post(&sync)
```

The Readers – Writers Problem

Description

- **Reading** data can happen **in parallel** without problems.
- **Writing needs mutual exclusion** to avoid race conditions.
- Aim it to exploit these observations to **efficiently synchronize** access to the data.

The Readers – Writers Problem

Description

- **Reading** data can happen **in parallel** without problems.
- **Writing needs mutual exclusion** to avoid race conditions.
- Aim it to exploit these observations to **efficiently synchronize** access to the data.
- **Different solutions** exist to the readers/writers problem
  - Solution 1: naive implementation with **limited parallelism**
  - Solution 2: **readers** receive **priority**: **no reader is kept waiting** - unless a writer already has access. Issue: **writers may starve**.
  - Solution 3: **writing** is performed **as soon as possibly**. Issue: **readers may starve**.

The Readers – Writers Problem

Solution 1: No Parallelism

```
void* reader(void* arg)
{
  while(1)
  {
    pthread_mutex_lock(&sync);
    printf("reading, counter now %d\n", counter);
    pthread_mutex_unlock(&sync);
  }
}

void* writer(void* writer)
{
  while(1)
  {
    pthread_mutex_lock(&sync);
    printf("writing, counter now %d\n", counter++);
    pthread_mutex_unlock(&sync);
  }
}
```

The Readers – Writers Problem

Solution 2: Readers First

- Solution 1: prevents **parallel reading**
- Solution 2: **allows parallel reading**
- A correct implementation of solution 2 requires:
  - `iReadCount`: an integer tracking the number of readers
    - If `iReadCount > 0`: writers are blocked (`sem_wait(rwSync)`)
    - If `iReadCount == 0`: writers are released (`sem_post(rwSync)`)
    - If already writing, readers must wait
  - `sync`: a mutex for mutual exclusion of `iReadCount`
  - `rwSync`: a semaphore that **synchronises the readers and writers**, set by the **first/last reader**

The Readers – Writers Problem

Solution 2: Readers First

```
void * reader(void * arg)
{
  while(1)
  {
    pthread_mutex_lock(&sync);
    iReadCount++;
    if(iReadCount == 1)
      sem_wait(&rwSync);
    pthread_mutex_unlock(&sync);

    printf("reading record\n");

    pthread_mutex_lock(&sync);
    iReadCount--;
    if(iReadCount == 0)
      sem_post(&rwSync);
    pthread_mutex_unlock(&sync);
  }
}
```

```
void * writer(void * writer)
{
  while(1)
  {
    sem_wait(&rwSync);
    printf("writing\n");
    sem_post(&rwSync);
  }
}
```

The Readers – Writers Problem

Solution 2: Readers First

```
void * reader(void * arg)
{
  while(1)
  {
    pthread_mutex_lock(&sync);
    iReadCount++;
    if(iReadCount == 1)
      sem_wait(&rwSync);
    pthread_mutex_unlock(&sync);

    printf("reading record\n");

    pthread_mutex_lock(&sync);
    iReadCount--;
    if(iReadCount == 0)
      sem_post(&rwSync);
    pthread_mutex_unlock(&sync);
  }
}
```

```
void * writer(void * writer)
{
  while(1)
  {
    sem_wait(&rwSync);
    printf("writing\n");
    sem_post(&rwSync);
  }
}
```

The Readers – Writers Problem

Solution 2: Readers First

```
void * reader(void * arg)
{
  while(1)
  {
    pthread_mutex_lock(&sync); // 1=>0
    iReadCount++;
    if(iReadCount == 1)
      sem_wait(&rwSync);
    pthread_mutex_unlock(&sync);

    printf("reading record\n");

    pthread_mutex_lock(&sync);
    iReadCount--;
    if(iReadCount == 0)
      sem_post(&rwSync);
    pthread_mutex_unlock(&sync);
  }
}
```

```
void * writer(void * writer)
{
  while(1)
  {
    sem_wait(&rwSync);
    printf("writing\n");
    sem_post(&rwSync);
  }
}
```

The Readers – Writers Problem

Solution 2: Readers First

```
void * reader(void * arg)
{
  while(1)
  {
    pthread_mutex_lock(&sync);
    iReadCount++;  // 0=>1
    if(iReadCount == 1)
      sem_wait(&rwSync);
    pthread_mutex_unlock(&sync);

    printf("reading record\n");

    pthread_mutex_lock(&sync);
    iReadCount--;
    if(iReadCount == 0)
      sem_post(&rwSync);
    pthread_mutex_unlock(&sync);
  }
}
```

```
void * writer(void * writer)
{
  while(1)
  {
    sem_wait(&rwSync);
    printf("writing\n");
    sem_post(&rwSync);
  }
}
```

The Readers – Writers Problem

Solution 2: Readers First

```
void * reader(void * arg)
{
  while(1)
  {
    pthread_mutex_lock(&sync);
    iReadCount++;
    if(iReadCount == 1)
      sem_wait(&rwSync);  // 1=>0
    pthread_mutex_unlock(&sync);

    printf("reading record\n");

    pthread_mutex_lock(&sync);
    iReadCount--;
    if(iReadCount == 0)
      sem_post(&rwSync);
    pthread_mutex_unlock(&sync);
  }
}
```

```
void * writer(void * writer)
{
  while(1)
  {
    sem_wait(&rwSync);
    printf("writing\n");
    sem_post(&rwSync);
  }
}
```

The Readers – Writers Problem

Solution 2: Readers First

```
void * reader(void * arg)
{
  while(1)
  {
    pthread_mutex_lock(&sync);
    iReadCount++;
    if(iReadCount == 1)
      sem_wait(&rwSync);
    pthread_mutex_unlock(&sync); // 0=>1

    printf("reading record\n");

    pthread_mutex_lock(&sync);
    iReadCount--;
    if(iReadCount == 0)
      sem_post(&rwSync);
    pthread_mutex_unlock(&sync);
  }
}
```

```
void * writer(void * writer)
{
  while(1)
  {
    sem_wait(&rwSync);
    printf("writing\n");
    sem_post(&rwSync);
  }
}
```

The Readers – Writers Problem

Solution 2: Readers First

```
void * reader(void * arg)
{
  while(1)
  {
    pthread_mutex_lock(&sync);
    iReadCount++;
    if(iReadCount == 1)
      sem_wait(&rwSync);
    pthread_mutex_unlock(&sync);

    printf("reading record\n");

    pthread_mutex_lock(&sync);
    iReadCount--;
    if(iReadCount == 0)
      sem_post(&rwSync);
    pthread_mutex_unlock(&sync);
  }
}
```

```
void * writer(void * writer)
{
  while(1)
  {
    sem_wait(&rwSync);   // 0=>-1
    printf("writing\n");
    sem_post(&rwSync);
  }
}
```

The Readers – Writers Problem

Solution 2: Readers First

```
void * reader(void * arg)
{
  while(1)
  {
    pthread_mutex_lock(&sync);
    iReadCount++;
    if(iReadCount == 1)
      sem_wait(&rwSync);
    pthread_mutex_unlock(&sync);

    printf("reading record\n");

    pthread_mutex_lock(&sync);
    iReadCount--;
    if(iReadCount == 0)
      sem_post(&rwSync);
    pthread_mutex_unlock(&sync);
  }
}
```

```
void * writer(void * writer)
{
  while(1)
  {
    sem_wait(&rwSync);
    printf("writing\n");
    sem_post(&rwSync);
  }
}
```

The Readers – Writers Problem

Solution 2: Readers First

```
void * reader(void * arg)
{
  while(1)
  {
    pthread_mutex_lock(&sync);
    iReadCount++;
    if(iReadCount == 1)
      sem_wait(&rwSync);
    pthread_mutex_unlock(&sync);

    printf("reading record\n");

    pthread_mutex_lock(&sync);  // 1=>0
    iReadCount--;
    if(iReadCount == 0)
      sem_post(&rwSync);
    pthread_mutex_unlock(&sync);
  }
}
```

```
void * writer(void * writer)
{
  while(1)
  {
    sem_wait(&rwSync);
    printf("writing\n");
    sem_post(&rwSync);
  }
}
```

The Readers – Writers Problem

Solution 2: Readers First

```
void * reader(void * arg)
{
  while(1)
  {
    pthread_mutex_lock(&sync);
    iReadCount++;
    if(iReadCount == 1)
      sem_wait(&rwSync);
    pthread_mutex_unlock(&sync);

    printf("reading record\n");

    pthread_mutex_lock(&sync);
    iReadCount--;  // 1=>0
    if(iReadCount == 0)
      sem_post(&rwSync);
    pthread_mutex_unlock(&sync);
  }
}
```

```
void * writer(void * writer)
{
  while(1)
  {
    sem_wait(&rwSync);
    printf("writing\n");
    sem_post(&rwSync);
  }
}
```

The Readers – Writers Problem

Solution 2: Readers First

```
void * reader(void * arg)
{
  while(1)
  {
    pthread_mutex_lock(&sync);
    iReadCount++;
    if(iReadCount == 1)
      sem_wait(&rwSync);
    pthread_mutex_unlock(&sync);

    printf("reading record\n");

    pthread_mutex_lock(&sync);
    iReadCount--;
    if(iReadCount == 0)
      sem_post(&rwSync);  // -1=>0
    pthread_mutex_unlock(&sync);
  }
}
```

```
void * writer(void * writer)
{
  while(1)
  {
    sem_wait(&rwSync);  // wakeup
    printf("writing\n");
    sem_post(&rwSync);
  }
}
```

The Readers – Writers Problem

Solution 2: Readers First

```
void * reader(void * arg)
{
  while(1)
  {
    pthread_mutex_lock(&sync);
    iReadCount++;
    if(iReadCount == 1)
      sem_wait(&rwSync);
    pthread_mutex_unlock(&sync);

    printf("reading record\n");

    pthread_mutex_lock(&sync);
    iReadCount--;
    if(iReadCount == 0)
      sem_post(&rwSync);
    pthread_mutex_unlock(&sync);  // 0=>1
  }
}
```

```
void * writer(void * writer)
{
  while(1)
  {
    sem_wait(&rwSync);
    printf("writing\n");
    sem_post(&rwSync);
  }
}
```

The Readers – Writers Problem

Solution 2: Readers First

```
void * reader(void * arg)
{
  while(1)
  {
    pthread_mutex_lock(&sync);
    iReadCount++;
    if(iReadCount == 1)
      sem_wait(&rwSync);
    pthread_mutex_unlock(&sync);

    printf("reading record\n");

    pthread_mutex_lock(&sync);
    iReadCount--;
    if(iReadCount == 0)
      sem_post(&rwSync);
    pthread_mutex_unlock(&sync);
  }
}
```

```
void * writer(void * writer)
{
  while(1)
  {
    sem_wait(&rwSync);
    printf("writing\n");
    sem_post(&rwSync);
  }
}
```

The Readers – Writers Problem

Solution 2: Readers First

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void * reader(void * arg)
{
  while(1)
  {
    pthread_mutex_lock(&sync);
    iReadCount++;
    if(iReadCount == 1)
      sem_wait(&rwSync);
    pthread_mutex_unlock(&sync);

    printf("reading record\n");

    pthread_mutex_lock(&sync);
    iReadCount--;
    if(iReadCount == 0)
      sem_post(&rwSync);
    pthread_mutex_unlock(&sync);
  }
}
```

```
void * writer(void * writer)
{
  while(1)
  {
    sem_wait(&rwSync);
    printf("writing\n");
    sem_post(&rwSync);   // 0=>1
  }
}
```

The Readers – Writers Problem

Solution 2: Readers First

```
void * reader(void * arg)
{
  while(1)
  {
    pthread_mutex_lock(&sync);
    iReadCount++;
    if(iReadCount == 1)
      sem_wait(&rwSync);
    pthread_mutex_unlock(&sync);

    printf("reading record\n");

    pthread_mutex_lock(&sync);
    iReadCount--;
    if(iReadCount == 0)
      sem_post(&rwSync);
    pthread_mutex_unlock(&sync);
  }
}
```

```
void * writer(void * writer)
{
  while(1)
  {
    sem_wait(&rwSync); // 1=>0
    printf("writing\n");
    sem_post(&rwSync);
  }
}
```

The Readers – Writers Problem

Solution 2: Readers First

```
void * reader(void * arg)
{
  while(1)
  {
    pthread_mutex_lock(&sync); // 1=>0
    iReadCount++;
    if(iReadCount == 1)
      sem_wait(&rwSync);
    pthread_mutex_unlock(&sync);

    printf("reading record\n");

    pthread_mutex_lock(&sync);
    iReadCount--;
    if(iReadCount == 0)
      sem_post(&rwSync);
    pthread_mutex_unlock(&sync);
  }
}
```

```
void * writer(void * writer)
{
  while(1)
  {
    sem_wait(&rwSync);
    printf("writing\n");
    sem_post(&rwSync);
  }
}
```

The Readers – Writers Problem

Solution 2: Readers First

```
void * reader(void * arg)
{
  while(1)
  {
    pthread_mutex_lock(&sync);
    iReadCount++; // 0=>1
    if(iReadCount == 1)
      sem_wait(&rwSync);
    pthread_mutex_unlock(&sync);

    printf("reading record\n");

    pthread_mutex_lock(&sync);
    iReadCount--;
    if(iReadCount == 0)
      sem_post(&rwSync);
    pthread_mutex_unlock(&sync);
  }
}
```

```
void * writer(void * writer)
{
  while(1)
  {
    sem_wait(&rwSync);
    printf("writing\n");
    sem_post(&rwSync);
  }
}
```

The Readers – Writers Problem

Solution 2: Readers First

```
void * reader(void * arg)
{
  while(1)
  {
    pthread_mutex_lock(&sync);
    iReadCount++;
    if(iReadCount == 1)
      sem_wait(&rwSync); // 0=>-1 (sleep)
    pthread_mutex_unlock(&sync);

    printf("reading record\n");

    pthread_mutex_lock(&sync);
    iReadCount--;
    if(iReadCount == 0)
      sem_post(&rwSync);
    pthread_mutex_unlock(&sync);
  }
}
```

```
void * writer(void * writer)
{
  while(1)
  {
    sem_wait(&rwSync);
    printf("writing\n");
    sem_post(&rwSync);
  }
}
```

The Readers – Writers Problem

Solution 2: Readers First

```
void * reader(void * arg)
{
  while(1)
  {
    pthread_mutex_lock(&sync);
    iReadCount++;
    if(iReadCount == 1)
      sem_wait(&rwSync);
    pthread_mutex_unlock(&sync);

    printf("reading record\n");

    pthread_mutex_lock(&sync);
    iReadCount--;
    if(iReadCount == 0)
      sem_post(&rwSync);
    pthread_mutex_unlock(&sync);
  }
}
```

```
void * writer(void * writer)
{
  while(1)
  {
    sem_wait(&rwSync);
    printf("writing\n");
    sem_post(&rwSync);
  }
}
```

The Readers – Writers Problem

Solution 2: Readers First

```
void * reader(void * arg)
{
  while(1)
  {
    pthread_mutex_lock(&sync);
    iReadCount++;
    if(iReadCount == 1)
      sem_wait(&rwSync);  (wakeup)
    pthread_mutex_unlock(&sync);

    printf("reading record\n");

    pthread_mutex_lock(&sync);
    iReadCount--;
    if(iReadCount == 0)
      sem_post(&rwSync);
    pthread_mutex_unlock(&sync);
  }
}
```

```
void * writer(void * writer)
{
  while(1)
  {
    sem_wait(&rwSync);
    printf("writing\n");
    sem_post(&rwSync);  // -1=>0
  }
}
```

The Readers – Writers Problem

Solution 2: Readers First

```
void * reader(void * arg)
{
  while(1)
  {
    pthread_mutex_lock(&sync);
    iReadCount++;
    if(iReadCount == 1)
      sem_wait(&rwSync);
    pthread_mutex_unlock(&sync);

    printf("reading record\n");

    pthread_mutex_lock(&sync);
    iReadCount--;
    if(iReadCount == 0)
      sem_post(&rwSync);
    pthread_mutex_unlock(&sync);
  }
}
```

```
void * writer(void * writer)
{
  while(1)
  {
    sem_wait(&rwSync);
    printf("writing\n");
    sem_post(&rwSync);
  }
}
```

The Readers – Writers Problem

Solution 2: Readers First

- Assuming a fair semaphore implementation, even with lots of writers, **readers will still get to run**.
- With lots of readers, **writers may starve** as the readers never post to `rwSync`.

Check you understanding

- Using our optimal parallelism dining philosophers solution, can a philosopher starve?
- Readers-writers seems to apply to any shared data, how could we decide when to use it?